

HARSHITA GUPTA

[E-mail](#) | [LinkedIn](#) | +1-(732)496-0919 | [GitHub](#)

EDUCATION

New Jersey Institute of Technology, New Jersey

Jan 2024-May 2025

Masters of Science in Computer Science (GPA: 3.75)

Indraprastha Institute of Information Technology, New Delhi

July 2019-June 2023

Bachelor of Science in Computer Science Engineering with Artificial Intelligence

WORK EXPERIENCE

Objectstream | Data Engineer

Jan 2024-Present

- Built and shipped a **GenAI-powered SQL assistant** using open-source LLMs, schema-aware prompting, and React UI — cut query drafting time by **80%**.
- Built **PySpark ETL pipelines** processing **40M+ records** with fault tolerance, improving throughput by **80%**.
- Developed **SQL-based analytics pipelines** (PostgreSQL, MongoDB, AWS S3), ensuring data reliability and availability.
- Python based data validation anomaly detection modules developed to enforce business logic, reducing QA overhead by **60%** and catching **95%** of onboarding data issues before production.
- Operated workloads in **AWS + GCP**, gaining hands-on experience in cloud-native development.
- Managed CI/CD infrastructure using GitHub, streamlining deployments and regression testing across development and production environments.

NKR Solutions | Software Developer

Nov 2023-Feb 2024

- Designed** and implemented trading algorithms (Three Black Crows, Breakout) with **ML (Random Forest, XGBoost)**, improving trade signal accuracy and **automating** 80% of manual processes.
- Built** modular broker API integrations (Zerodha, Interactive Brokers) with authentication & testing, enhancing system reliability.
- Built real-time trading automation in Python, improving execution speed and automating **80%** of manual workflows.
- Created **Python** automation scripts for strategy validation, reducing QA cycles by **50%**.

Indraprastha Institute of Information Technology | Research Assistant

Sep 2023-Jan 2024

- Designed an AI-based **human depression detection system** with multimodal deep learning, combining stacked **LSTMs** and **SHAP** analysis; submitted findings to the **ACII** conference.
- Built and pretrained a Multi-Modal Masked Autoencoder on the Kinetics dataset for action video reconstruction, improving representation accuracy across RGB, Depth, and Semantic modalities.

Bluekactus | Software Engineer Intern

May 2022-Aug 2022

- Optimized SQL procedures and backend APIs for inventory modules, reducing query time by **40%**.

Cadence Design Systems | Summer Trainee

May 2021-Aug 2021

- Built **frontend single-page apps** (PHP, JavaScript, HTML/CSS) to automate simulation comparisons, improving engineer productivity.

TECHNICAL SKILLS

Languages & Frameworks: Python, Java, C++, HTML, SQL, Data Structure, OOP

Tools/Technologies: AWS (EC2, S3), GCP, Google Kubernetes Engine (GKE), Docker, Linux, Agile/Scrum, GitHub Actions (CI/CD), PySpark, Airflow, Kafka, PostgreSQL, MongoDB, Cassandra, Computer Vision (YOLOv5, OpenCV, CUDA), NLP, Pandas, NumPy, PyTorch, TensorFlow, HuggingFace, Transformers, LLM fine-tuning

SELECTED PROJECTS

GenAI-Powered Slide QA System:

- Built a **RAG** pipeline to answer natural language questions over PDF slide decks by combining semantic retrieval (SentenceTransformers + FAISS) with LLM-based answer generation (**FLAN-T5**).
- Designed modular Python codebase with CI/CD pipelines and automated testing for reliability.

Image captioning with CNN-LSTM and attention:

- Implemented an image captioning model using **ResNet-101**, **Bahdanau attention**, and **LSTM** decoder to generate fluent, context-aware captions, achieving a BLEU-4: 0.1637 on Flickr8k using **TensorFlow** and beam search decoding.
- Built using **Python**, **TensorFlow**, **Keras**, **Wulver**, **NLTK**, **Matplotlib**

Distracted Driver Detection Real Time System:

- Developed YOLOv5-based real-time vehicle detection optimized for traffic cameras. Used: **Python**, **PyTorch**, **OpenCV**, **CUDA**, **YOLOv5**
- Reduced inference time and model size using GPU acceleration and parallelization techniques.

Stock market prediction using RL:

- Pioneered Safety-Critic and Multi-Agent techniques within a dynamic RL pipeline (**PPO**, **A2C**, **DDPG**), boosting Sharpe ratio by **30%** over ensemble baseline.
- Built and trained models using **Python**, **PyTorch**, **Stable-Baselines3**, and Gymnasium; deployed on NJIT's **Wulver HPC cluster**.

ACHIEVEMENTS

- Dean's List for Academic Excellence, 2020-2021
- Google Cloud badge: Introduction to Generative AI
- Mentor: Student Mentorship Program, 2022-2023, guided 15+ junior students
- Winner: Game Design Hackathon, 2021, 1st place among 20+ teams